# An Echo State Network with a Ridge Readout on Filter-Bank CSP Signals for Motor-Imagery EEG

MI-EEG homework report · BCI Competition IV-2a · comparison with MixNet and FBCSP+SVM

**Abstract.** Reservoir computing proposes that a fixed, random recurrent network turns an input into a rich state from which a simple linear readout can learn, and recurrent cortical circuits have been described in these terms. I test the idea on two-class motor-imagery EEG. The preprocessing produces 18 filter-bank CSP signals, normalised within each trial and band to reduce day-to-day change. The model is an echo state network (ESN): a 54-unit random reservoir that is never trained, log-variance pooling of the unit states, and a ridge-regression readout solved in closed form, with 110 trained parameters. The reservoir settings and the ridge strength were found by CMA-ES on validation folds only. Under the MixNet protocol (9 subjects, 5-fold cross-validation in session T, session E as test) the ESN reaches 78.27 ± 14.43 %, against 77.59 ± 15.24 % for MixNet (44,831 parameters) and 78.27 ± 15.40 % for FBCSP+SVM; neither difference is significant. The leave-one-out comparison of the new modules shows that per-trial normalisation and standardisation matter most (2.2 and 3.1 test points when removed), and that the reservoir helps only slightly (0.8 test points, 1.9 validation points). The errors of all three models are lopsided within subjects, which is consistent with a shift of the decision threshold between days.

## 1 Introduction

Imagining a hand movement lowers mu (8–13 Hz) and beta (13–30 Hz) power over the motor cortex opposite to the hand (event-related desynchronisation). The information therefore lies in the power of oscillations, in subject-specific bands and channel combinations. BCI Competition IV-2a [3] offers 144 trials per session for the two classes used here, so a cross-validation fold has about 115 training trials, and the test session E is recorded on a different day from the training session T. Two problems follow: models must be small enough for the data, and features must survive the change of day.

An echo state network [5, 6] answers the first problem by construction. Its recurrent weights are drawn at random and never trained; the input drives this reservoir, whose state mixes the recent input history, and only a linear readout of the state is learned, classically by ridge regression [11]. The idea is close to a view of cortical microcircuits as reservoirs whose transient dynamics are read out downstream [7]. MixNet [2], the baseline, takes the opposite route: a convolutional encoder trained with three losses.

**Hypotheses.** (H1) A pipeline whose only trained network part is a linear readout of a random reservoir is competitive with MixNet. (H2) The reservoir adds information: the same readout without it is worse. (H3) Normalising each band within each trial makes the signals more robust to the change of recording day. Accuracy results cannot show how the brain computes; they only show whether a random recurrent network is a useful feature extractor for these signals.

## 2 Method

### 2.1 Overview

Figure 1 compares the two pipelines. Both start with a filter bank of 9 bands (4–40 Hz in 4 Hz steps, as in FBCSP [1]) and 2 CSP filters per band [4], which gives 18 signals of 400 samples per trial. MixNet then learns temporal features with a convolutional encoder and three training losses. My pipeline normalises the signals, drives a fixed random reservoir with them, pools the power of every unit over time and reads the result out with ridge regression. Modules tagged "new" are not in MixNet.

For comparison, MixNet is trained end to end on the loss of Eq. (1): a reconstruction error of its decoder, a semi-hard triplet loss on the latent vector z (margin m = 1, d the Euclidean distance, n a negative trial of the other class), and cross-entropy on the class probabilities p, with weights ω adapted during training by gradient blending. All 44,831 of its parameters follow this loss. In my pipeline nothing is trained by gradient descent.

$$\mathcal{L}_{\text{MixNet}} = \omega_1 \frac{1}{|\mathbf{X}|}\|\mathbf{X}-\hat{\mathbf{X}}\|^2 + \omega_2 \frac{1}{|\mathcal{P}|}\sum_{(a,p)\in\mathcal{P}} \max(0,\, d(\mathbf{z}_a,\mathbf{z}_p) - d(\mathbf{z}_a,\mathbf{z}_n) + m) - \omega_3 \frac{1}{n}\sum_{i=1}^{n} \log p_{i,c_i} \qquad (1)$$

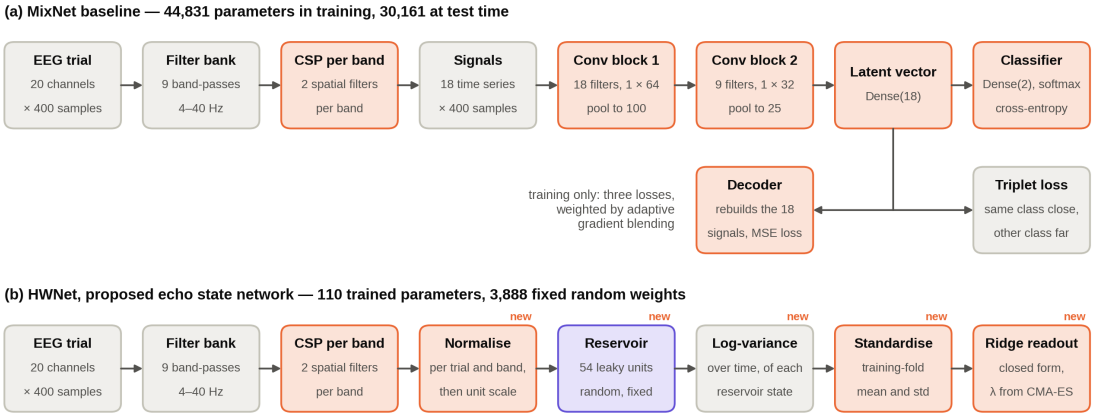



**Figure 1.** The two pipelines. (a) MixNet baseline. (b) Proposed pipeline. Orange: fitted on the training fold; violet: random and never trained; grey: fixed operation.

## 2.2 Modules and why each is needed

Table 1 lists every module with its purpose. The last column anticipates Section 4.3: the change in validation and test accuracy when only that module is removed (for the direct inputs, which the final model does not use: when they are added).

**Table 1.** Modules of the proposed pipeline. Last column: change in validation / test accuracy (points) when only that module is removed from the final model (final: 82.0 / 78.3 %).

| Module | What it does | Why it is needed | Removed: val / test |
|---|---|---|---|
| Filter bank | 9 zero-phase Butterworth band-passes, 4–40 Hz | The reactive band differs between subjects; separate bands let CSP find band-specific patterns. | shared |
| CSP per band | 2 spatial filters per band from trace-normalised class covariances (standard FBCSP front end) | Combines the 20 channels so that power differs most between the two classes. | shared |
| Per-trial band normalisation (new) | Divides each band's 2 signals by their RMS in the same trial | Overall band power drifts between days; the balance between the 2 CSP filters carries the class. One trial at a time, so it cannot leak. | +0.2 / −2.2 |
| Unit scaling (new) | One scale per signal from the training fold | Gives the reservoir input a fixed scale across subjects and folds. | +0.2 / −0.1 |
| Reservoir (new) | 54 leaky tanh units, random sparse recurrence; never trained | The hypothesis: mixes the recent history of the 18 signals at no training cost. | −1.9 / −0.8 |
| Log-variance pooling (new) | Log of the variance over time of each unit state, after a 0.5 s washout | Class information is in power; the mean of an oscillation is about zero. | −30.3 / −27.4 |
| Standardisation (new) | Training-fold mean and standard deviation of each pooled value | Makes the ridge penalty treat all 54 values alike. | −3.4 / −3.1 |
| Ridge readout (new) | Closed-form ridge regression to ±1 targets, λ = 64; 110 parameters | The only trained part; closed form avoids the noise of gradient training on ~115 trials. | — |
| Direct inputs (tested, not used) | Also feed the log-variance of the 18 input signals to the readout (+18 units of readout input) | Standard ESN option [6]. It gave no gain here, so the smaller model is kept (Section 4.3). | −0.2 / −0.2 |

### 2.3 What each module outputs

Figure 2 follows two validation trials of subject A03 (one left-hand, one right-hand) through every module; all fitted parts come from the training fold of the same split. Row 1 is the raw EEG at C3 and C4, over the left and right motor cortex. Row 2 keeps only 8–12 Hz, where the mu rhythm becomes visible. Row 3 shows the two CSP filters of that band: in the left-hand trial filter 2 carries more power than filter 1 (variance 217 against 58), in the right-hand trial filter 1 more than filter 2 (284 against 114), which is exactly the contrast CSP is built to produce. Row 4 is the same pair after per-trial normalisation and unit scaling: the shape is unchanged, the overall level is removed and only the balance between the filters remains. Row 5 shows the 54 reservoir states driven by all 18 signals; they are small (RMS 0.024, largest value 0.14, colour scale adapted to each trial), and the first 0.5 s (washout) is discarded because the state still depends on its starting point. Row 6 is the pooled, standardised log-variance of the 54 units, the vector the readout sees. Row 7 is the readout output for all validation trials of this fold: the sign of the score difference gives the class, and 28 of 29 trials fall on the correct side.

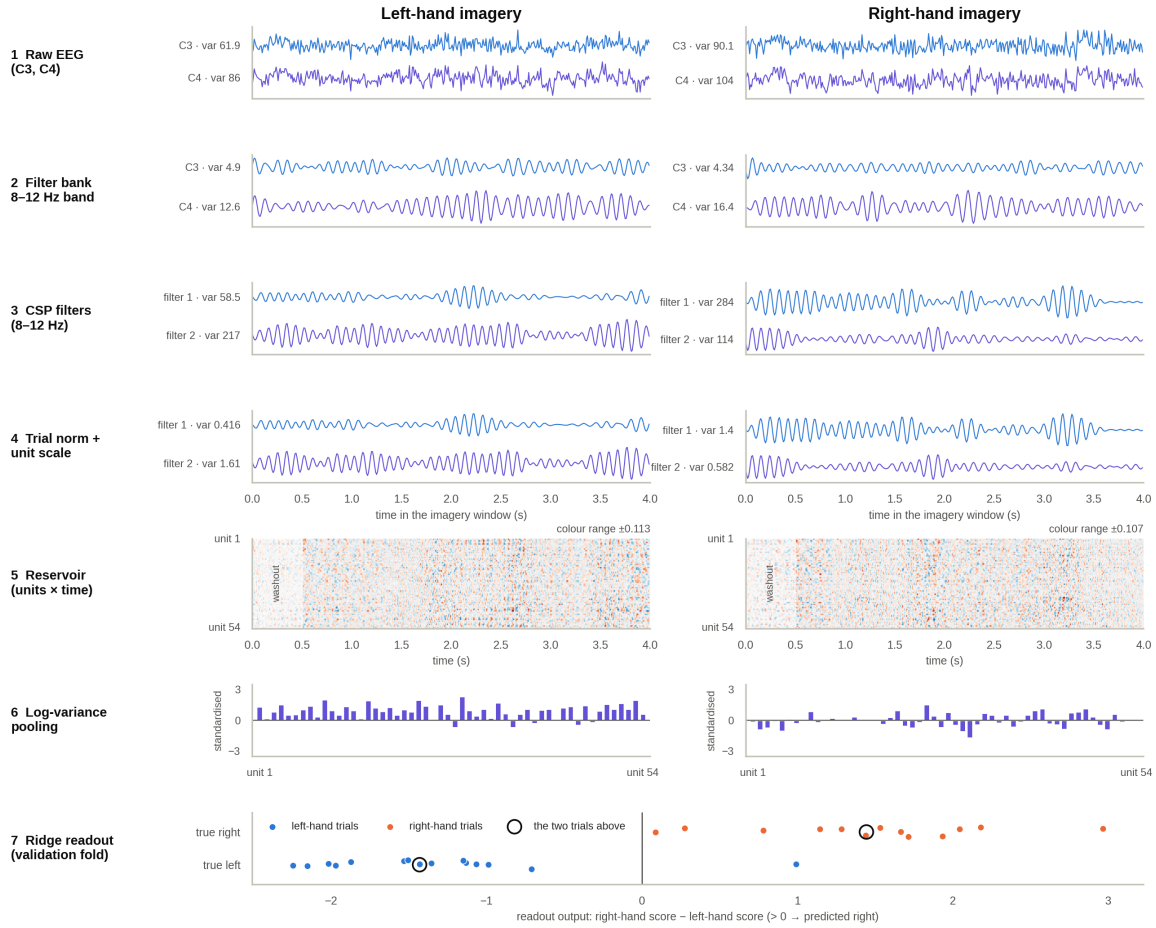



**Figure 2.** Output of every module for one left-hand and one right-hand validation trial of subject A03 (fold 1); row 7 shows the readout output for all validation trials of that fold.

### 2.4 The pipeline in equations

A trial is X in $R^{C\times T}$ with C = 20 channels and T = 400 samples (4 s at $f_s$ = 100 Hz); i indexes trials, c in {0, 1} the class (left, right hand) and b = 1, …, 9 the frequency band. Every quantity marked as learned is computed from the training trials of the current fold only.

**Filter bank.** Each band applies a 4th-order Butterworth band-pass $H_b$ forward and then backward in time (Eq. 2). The combined response $G_b$ has the squared magnitude of $H_b$ and zero phase, so the oscillations keep their timing. Nothing is learned.

$$\mathbf{X}_b = \overleftarrow{H}_b(H_b(\mathbf{X})), \qquad |G_b(f)| = |H_b(f)|^2, \quad \angle G_b(f) = 0, \qquad [f_b^{\text{lo}}, f_b^{\text{hi}}] = [4b,\, 4b+4]\ \text{Hz} \tag{2}$$

**CSP per band (learned).** Eq. (3) forms the trace-normalised covariance of every training trial and averages it per class ($I_c$ are the training trials of class c). Eq. (4) is the generalised eigenproblem of CSP: the eigenvalue λ(w) is the share of the power of the filtered signal that belongs to class 0, so the eigenvectors with the smallest and largest λ are the two filters whose output power differs most between the classes. Projecting the band-passed trial $x_b(t)$ in $R^{20}$ on them gives $z_b(t)$ in $R^2$; the 9 bands give 18 signals.

$$\mathbf{C}_i = \frac{\mathbf{X}_{b,i}\mathbf{X}_{b,i}^{\top}}{\mathrm{tr}(\mathbf{X}_{b,i}\mathbf{X}_{b,i}^{\top})}, \qquad \bar{\mathbf{C}}_c = \frac{1}{|I_c|}\sum_{i \in I_c} \mathbf{C}_i \tag{3}$$

$$\bar{\mathbf{C}}_0\,\mathbf{w} = \lambda(\bar{\mathbf{C}}_0 + \bar{\mathbf{C}}_1)\,\mathbf{w} \quad \Leftrightarrow \quad \lambda(\mathbf{w}) = \frac{\mathbf{w}^{\top}\bar{\mathbf{C}}_0\mathbf{w}}{\mathbf{w}^{\top}(\bar{\mathbf{C}}_0 + \bar{\mathbf{C}}_1)\mathbf{w}}, \qquad \mathbf{z}_b(t) = \mathbf{W}_b^{\top}\mathbf{x}_b(t), \;\; \mathbf{W}_b = [\mathbf{w}_{\min}, \mathbf{w}_{\max}] \tag{4}$$

**Per-trial band normalisation.** Eq. (5) divides the two CSP signals of band b by their joint root-mean-square in the same trial. A change in overall band power, such as a different electrode impedance on another day, multiplies both signals by the same factor and cancels; the ratio between the two filters, which carries the class, is kept. It uses one trial at a time, so it learns nothing.

$$\tilde{z}_{b,k}(t) = \frac{z_{b,k}(t)}{\sqrt{\frac{1}{2T}\sum_{k'=1}^{2}\sum_{t'=1}^{T} z_{b,k'}(t')^2}}, \qquad k = 1, 2 \tag{5}$$

**Unit scaling (learned).** Eq. (6) scales each of the 18 signals by its standard deviation over all training trials and samples ($m_j$ its mean), so the reservoir input u(t) in $R^{18}$ has unit scale.

$$u_j(t) = \frac{\tilde{z}_j(t)}{\sigma_j}, \qquad \sigma_j^2 = \frac{1}{nT}\sum_{i \in \mathrm{train}}\sum_{t=1}^{T}(\tilde{z}_{i,j}(t) - m_j)^2, \qquad j = 1, \ldots, 18 \tag{6}$$

**Reservoir (random, never trained).** Eq. (7) updates the state x(t) in $R^{N}$ of N = 54 units at every sample: the new input and the previous state pass through tanh, and the leak a = 0.75 mixes the result with the previous state, which slows the dynamics. Eq. (8) draws the fixed weights once from a seeded generator: input weights V in $R^{N\times 18}$ uniform with scale s = 0.012, and a sparse recurrent matrix (density p = 0.32) rescaled so that its spectral radius equals ρ = 0.34. Because ρ < 1, the influence of the starting state fades and x(t) depends only on the recent input history (the echo state property). V and W hold 3,888 fixed weights, the same for every subject and fold.

$$\mathbf{x}(t) = (1-a)\,\mathbf{x}(t-1) + a\tanh(\mathbf{V}\mathbf{u}(t) + \mathbf{W}\mathbf{x}(t-1)), \qquad \mathbf{x}(0) = \mathbf{0} \tag{7}$$

$$V_{jk} \sim \mathcal{U}(-s, s), \qquad \tilde{W}_{jk} = \xi_{jk}\beta_{jk}, \;\; \xi_{jk} \sim \mathcal{U}(-1, 1), \;\; \beta_{jk} \sim \mathrm{Bernoulli}(p), \qquad \mathbf{W} = \rho\frac{\tilde{\mathbf{W}}}{\varrho(\tilde{\mathbf{W}})} \tag{8}$$

**Operating regime.** The search chose a small input scale (s = 0.012, next to the lower end of its range, 0.01), and the unit states stay small (RMS 0.024, largest value 0.14). In that range tanh is almost linear, so the reservoir behaves as a bank of random linear filters of the 18 input signals (Eq. 9, in the z domain), not as a strongly non-linear dynamical system. The log-variance of each unit is then a weighted band-power measure of the CSP signals with its own frequency response. This is my reading of the numbers, not a tested claim, and it limits what the result says about non-linear reservoirs.

$$\mathbf{X}(z) \approx a[(z - 1 + a)\mathbf{I} - a\mathbf{W}]^{-1}\mathbf{V}\,\mathbf{U}(z), \qquad \tanh(v) \approx v \;\text{ for }\; |v| \ll 1 \tag{9}$$

**Log-variance pooling.** Eq. (10) reduces each trial to one number per unit: the log of its variance after a washout of $T_0$ = 50 samples (0.5 s), with ε = $10^{-8}$, giving f in $R^{54}$. The variance measures band power, where the class information lies; the log makes it closer to Gaussian and turns ratios between filters into differences that a linear readout can use.

$$f_j = \log\left(\frac{1}{T - T_0}\sum_{t=T_0+1}^{T}(x_j(t) - \bar{x}_j)^2 + \varepsilon\right), \qquad j = 1, \ldots, N \tag{10}$$

**Standardisation (learned).** Eq. (11) centres and scales every pooled value with training-fold statistics, so that the ridge penalty below treats all 54 values alike.

$$\hat{f}_{i,j} = \frac{f_{i,j} - \mu_j}{\sigma_j^f}, \qquad \mu_j = \frac{1}{n}\sum_{i \in \mathrm{train}} f_{i,j}, \quad \left(\sigma_j^f\right)^2 = \frac{1}{n}\sum_{i \in \mathrm{train}}(f_{i,j} - \mu_j)^2 \tag{11}$$

**Ridge readout (learned).** The standardised features of the n training trials form the rows of **F**; A appends a column of ones for the bias, and the targets Y are +1 for the true class and −1 for the other (Eq. 12). Eq. (13) is ridge regression [11]: the least-squares fit plus a penalty λ = 64 on the weights (D leaves the bias unpenalised), solved exactly by one linear system, with no learning rate, epochs or early stopping. Eq. (14) gives the class scores of a new trial, the predicted class and the probabilities the softmax layer reports. Eq. (15) folds the standardisation into the output weights, so the saved network maps the input signals directly to scores; the trained parameters are the 110 entries of $\mathrm{W}_{\mathrm{out}}$ and b.

$$\mathbf{A} = [\hat{\mathbf{F}},\, \mathbf{1}], \qquad Y_{ic} = +1 \text{ if } c = c_i, \;\; -1 \text{ otherwise}, \qquad \mathbf{D} = \mathrm{diag}(1, \ldots, 1, 0) \tag{12}$$

$$\Theta^{*} = \arg\min_{\Theta} \|\mathbf{A}\Theta - \mathbf{Y}\|_F^2 + \lambda\,\|\mathbf{D}^{1/2}\Theta\|_F^2 = (\mathbf{A}^{\top}\mathbf{A} + \lambda\mathbf{D})^{-1}\mathbf{A}^{\top}\mathbf{Y}, \qquad \Theta = [\mathbf{W}_{\mathrm{out}};\; \mathbf{b}^{\top}] \tag{13}$$

$$\mathbf{g}(\mathbf{X}) = \mathbf{W}_{\text{out}}^{\top}\hat{\mathbf{f}} + \mathbf{b}, \qquad \hat{c} = \arg\max_{c}\, g_c(\mathbf{X}), \qquad p_c = \frac{e^{g_c}}{\sum_{c'} e^{g_{c'}}} \tag{14}$$

$$\mathbf{W}' = \operatorname{diag}(\sigma^f)^{-1}\,\mathbf{W}_{\text{out}}, \qquad \mathbf{b}' = \mathbf{b} - \mathbf{W}_{\text{out}}^{\top}\operatorname{diag}(\sigma^f)^{-1}\mu, \qquad |\theta_{\text{trained}}| = 2N + 2 = 110 \tag{15}$$

### 2.5 Hyperparameter search with CMA-ES

Because the reservoir is fixed and the readout is closed-form, one candidate setting can be scored in seconds without TensorFlow: the reservoir is run in NumPy on the training and validation trials of every fold, the readout is fitted on the training fold and the validation mean squared error is recorded. The objective (Eq. 16) is that error averaged over 3 random reservoirs r, 9 subjects s and 5 folds k, so the search cannot exploit one lucky draw; $V_{sk}$ is the validation fold and $g^{(\theta,r)}_{sk}$ the model fitted on the matching training fold. CMA-ES [10] (Eq. 17) samples 12 settings per generation from a Gaussian with mean m, step size σ and covariance C, moves m to a weighted mean of the 6 best ($\theta_{k:12}$ is the k-th best), and adapts σ and C from the successful steps. The six searched settings are the reservoir size (8–64), spectral radius (0–1.2), input scale (0.01–3.2), leak (0.05–1), connectivity (0.05–1) and ridge λ (0.001–1000), searched in [0, 1] and mapped to their ranges; the search ran 25 generations (300 settings), and session E was never loaded.

$$J(\theta) = \frac{1}{3\cdot 45}\sum_{r=1}^{3}\sum_{s=1}^{9}\sum_{k=1}^{5}\frac{1}{2|\mathcal{V}_{sk}|}\sum_{i\in\mathcal{V}_{sk}}\left\|\mathbf{y}_i - \mathbf{g}_{sk}^{(\theta,r)}(\mathbf{X}_i)\right\|^2, \qquad \theta = (N,\rho,s,a,p,\lambda) \tag{16}$$

$$\theta_k = \mathbf{m} + \sigma\,\mathcal{N}(\mathbf{0},\mathbf{C}), \quad k = 1,\ldots,12, \qquad \mathbf{m} \leftarrow \sum_{k=1}^{6} w_k\,\theta_{k:12} \tag{17}$$

Figure 3 shows the run: validation error fell from 0.778 for my hand-picked starting setting to 0.554. I ran the search twice. With the pooled input signals also fed to the readout (the earlier design) the best error was 0.563 with 138 parameters; without them it is 0.554 with 110 parameters. I had fixed in advance that I would keep the smaller model unless it was more than 0.01 worse; it is better, so the final model has no direct inputs. Smaller reservoirs are noticeably worse here: 18 units (38 parameters) reach 79.3 % validation accuracy against 81.7 % for the selected 54.

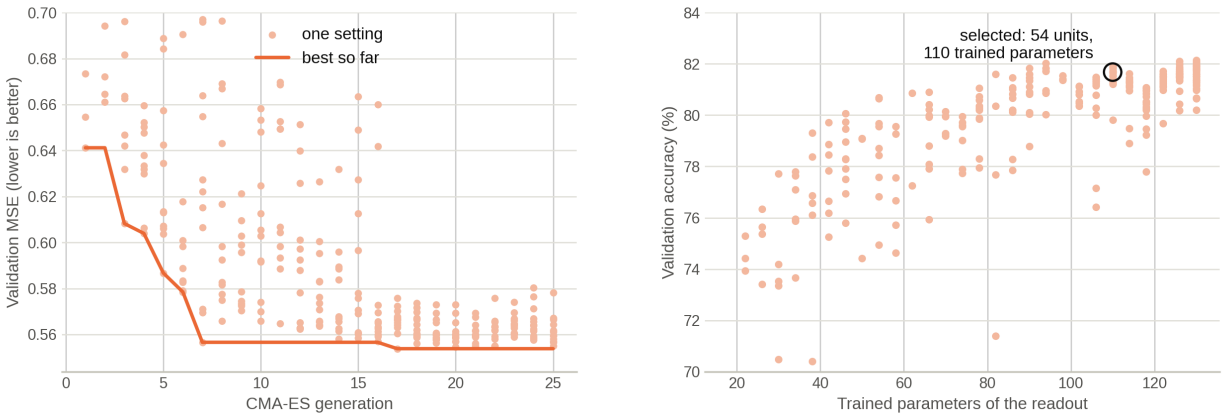



**Figure 3.** CMA-ES on validation folds. Left: validation MSE of every setting and the best so far, per generation. Right: validation accuracy against the number of trained readout parameters; the circle is the selected setting.

### 2.6 Leakage control

Data-driven steps: the CSP filters (with labels), the unit scale, the standardisation statistics and the ridge readout. All are fitted on the training fold only: the first two inside `fit_transform()`, stored in a state object and only read by `transform()`; the last two inside the model's `fit()`, which scores the validation fold without fitting on it. The per-trial normalisation uses one trial at a time and the reservoir is random. An automated test runs the real cross-validation driver and confirms that replacing session E, or the validation trials, with noise leaves every fitted state and all other outputs bit-identical, that a trial is transformed identically alone or in a batch, and that the CSP variants of Section 4.4 behave the same. All five tests pass. Hyperparameters were chosen on validation folds only.

## 3 Experimental setup

All models follow one protocol: 9 subjects, left vs. right hand, one model per subject and fold, stratified 5-fold cross-validation inside session T, session E (144 trials) as the test set of every fold, and the mean over folds within a subject, then over subjects. MixNet uses its published subject-dependent settings unchanged. FBCSP+SVM is the repository baseline: the same filter bank, 2 CSP filters per band, 8 features chosen by mutual information and an SVM whose kernel and C are chosen on the validation fold. Paired t-tests and Wilcoxon tests over the 9 subjects compare the models.

Eq. (18) defines the metrics: accuracy on the 144 trials of session E, macro F1 (the mean over the two classes of the per-class F1), and the subject score as the mean over the 5 folds. Eq. (19) defines the paired tests between two models A and B: $d_s$ is the per-subject difference, the t statistic has 8 degrees of freedom, and the Wilcoxon statistic W uses the ranks $R_s$ of $|d_s|$.

$$\mathrm{Acc} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}[\hat{c}_i = c_i], \qquad \mathrm{F1} = \frac{1}{2}\sum_{c=0}^{1} \frac{2\,\mathrm{TP}_c}{2\,\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c}, \qquad \mathrm{Acc}_s = \frac{1}{5}\sum_{k=1}^{5} \mathrm{Acc}_{s,k} \tag{18}$$

$$d_s = \mathrm{Acc}_s^A - \mathrm{Acc}_s^B, \qquad t = \frac{\bar{d}}{s_d/\sqrt{9}}, \qquad W = \min\left(\sum_{d_s>0} R_s, \sum_{d_s<0} R_s\right) \tag{19}$$

**How the design was reached.** Table 4 lists the designs I built in order. An attention network on windowed CSP features came first. I then replaced it with an ESN whose readout was trained by gradient descent, chose its reservoir in 22 validation-only configurations, and added the per-trial normalisation and CSP shrinkage. With that readout the reservoir did not help. I then replaced the readout by closed-form ridge regression, searched the settings with CMA-ES, and returned to plain CSP as in standard FBCSP: shrinkage variants had not been separable on validation, and plain CSP keeps the front end standard. A first version of the ridge model also fed the pooled input signals to the readout; the ablation showed no benefit, and I removed them and searched again (Section 2.5). Those choices were made before any test result of the final model was read. The ablations, CSP variants and extra reservoirs were trained afterwards and their test accuracies read only after the final model was fixed.

## 4 Results

### 4.1 Against MixNet and FBCSP+SVM

Figure 4 shows session E accuracy per subject for the three models and Table 2 the means. The three models rise and fall together: A03, A07 and A08 are easy for all, A02, A04 and A06 hard for all, and the spread between subjects (about 40 points) dwarfs the gaps between models. HWNet is +0.7 points above MixNet (paired t-test p = 0.71, Wilcoxon p = 1.00; for F1, p = 0.61 and 1.00), but ahead on only 3 of 9 subjects (A05, A06, A07): the higher mean comes mainly from A05, where HWNet gains 12 points. It is level with FBCSP+SVM (difference +0.0 points, p = 1.00, Wilcoxon p = 0.65; ahead on 6 of 9 subjects). FBCSP+SVM is itself level with MixNet, so on this dataset neither the convolutional network nor the reservoir beats the classical pipeline by a margin that nine subjects can show.

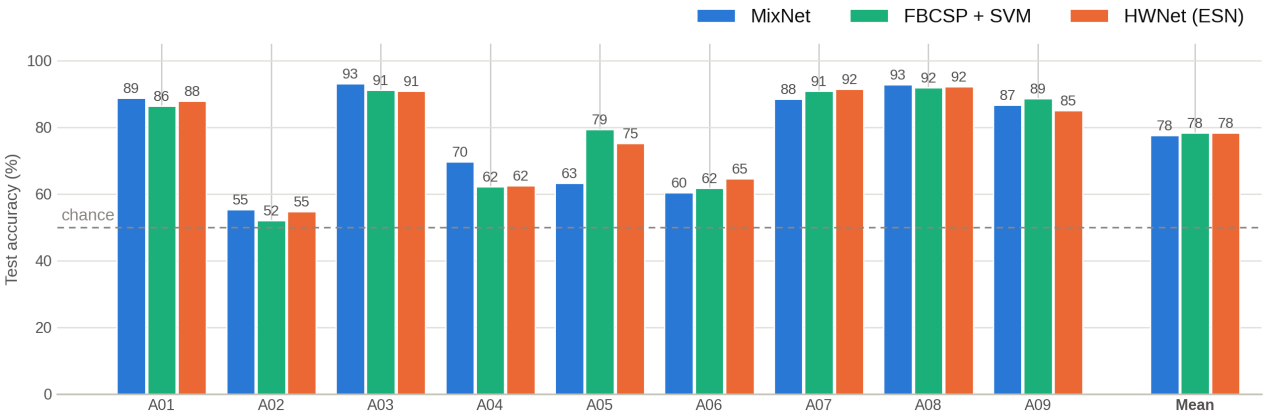



**Figure 4.** Session E accuracy per subject (mean of 5 folds) for MixNet, FBCSP+SVM and HWNet, and the mean over the 9 subjects.

**Table 2.** Mean ± standard deviation over the 9 subjects on session E.

| Model | Trained parameters | Test accuracy | Macro F1 |
|---|---|---|---|
| MixNet (baseline) | 44,831 (30,161 at test) | 0.7759 ± 0.1524 | 0.7662 ± 0.1629 |
| FBCSP + SVM (baseline) | SVM only | 0.7827 ± 0.1540 | 0.7751 ± 0.1632 |
| HWNet (proposed ESN) | 110 | 0.7827 ± 0.1443 | 0.7776 ± 0.1502 |

### 4.2 The change of recording day

Figure 5, left, sorts the per-subject differences to MixNet: HWNet's largest gain is on A05 and its largest loss on A04 (−7.2). Right: each model's mean validation accuracy (training day) joined to its mean test accuracy (another day). MixNet is better on validation (84.3 % against 82.0 %) but loses 6.7 points across days, HWNet 3.7. The smaller drop is consistent with H3, but a drop that is smaller by 3 points on nine subjects is only suggestive.

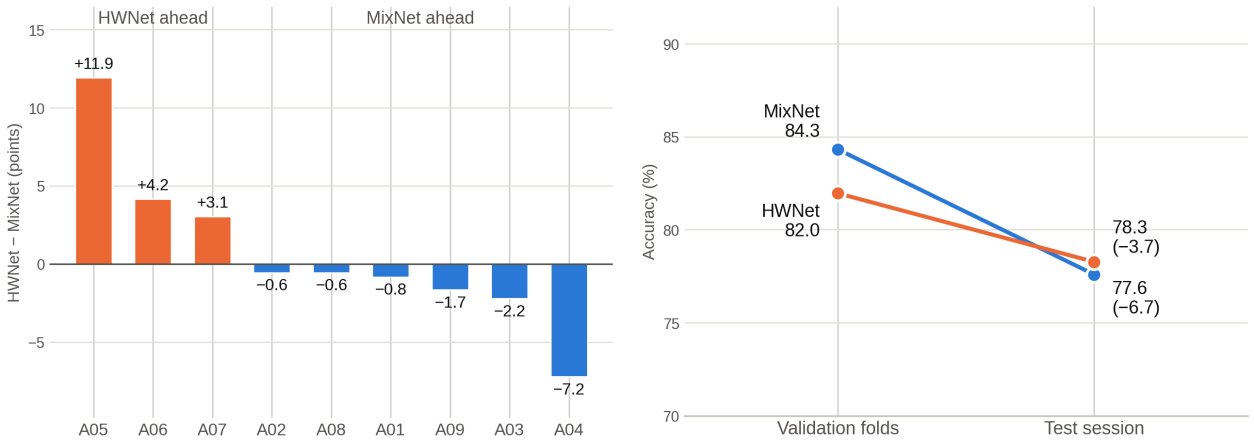



**Figure 5.** Left: accuracy difference HWNet − MixNet per subject, in points. Right: mean accuracy on the validation folds and on session E; the number in brackets is the change.

### 4.3 Comparing every new module

Figure 6 removes one new module at a time from the final model (Table 1, last column). Three random reservoirs give 78.3, 77.8, 78.3 %, so differences below about 0.4 points are within the reservoir's own randomness.

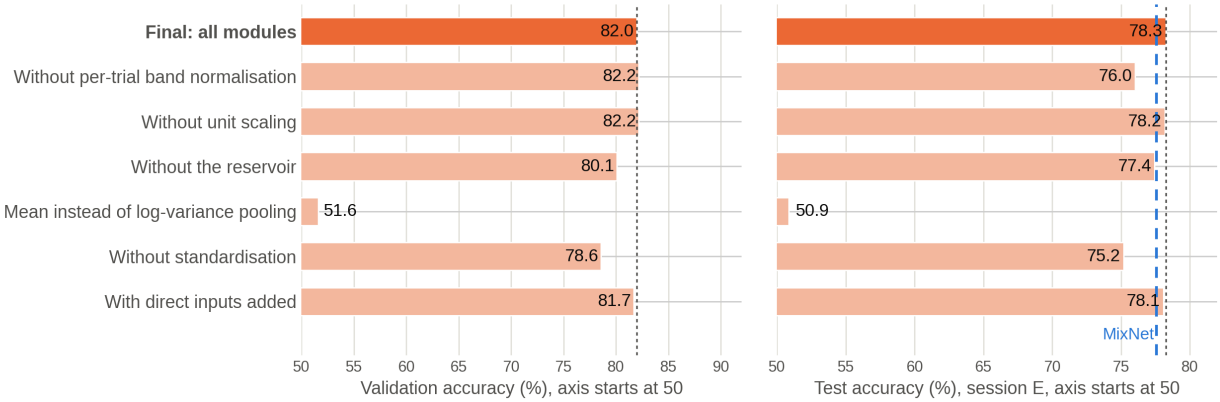



**Figure 6.** The final model with one change at a time. Left: validation accuracy; right: session E accuracy (dashed blue: MixNet). The last row adds the direct inputs, which the final model does not use. Both axes start at 50 %.

• **Log-variance pooling is essential.** With the mean over time instead, accuracy falls to 50.9 % on test and 51.6 % on validation, chance level: the mean of an oscillation carries no class information.
• **Standardisation helps clearly.** Without it test accuracy falls by 3.1 points to 75.2 % and validation by 3.4 points; the validation MSE rises from 0.548 to 0.810. With the small input scale the pooled values of different units differ strongly in spread, and the ridge penalty then treats them unequally (my explanation, not tested).
• **Per-trial normalisation helps (H3).** Removing it costs 2.2 test points, larger than the reservoir spread, while validation accuracy changes by +0.2 points and validation MSE rises from 0.548 to 0.581. The effect is much larger on the other day than on the training day, which is the pattern H3 predicts.
• **The reservoir helps slightly (H2, weakly).** Without it the same readout on the 18 pooled inputs reaches 77.4 % on test (−0.8 points) and 80.1 % on validation (−1.9 points), validation MSE 0.609. The test loss is only slightly above the reservoir spread; the validation loss is clearer.
• **Unit scaling makes no difference** (−0.1 test points).
• **Direct inputs add nothing.** Adding them changes test accuracy by −0.2 and validation accuracy by −0.2 points and raises the validation MSE to 0.564, so the smaller model is kept.

### 4.4 Other CSP variants

Table 3 trains the same network on four CSP variants, each with and without the reservoir. The reservoir gain is positive on test for all four, but small (0.1 to 1.1 points) and below the reservoir spread for two of them; on validation it is larger (0.5 to 2.7 points). Ledoit-Wolf shrinkage estimates almost no shrinkage on this data (mean intensity 0.001), so it behaves like plain CSP. A fixed 5 % shrinkage gives the best test accuracy of the four. Stationary CSP has the best validation accuracy and the worst test accuracy: it stabilises power across the trials of the training day, not across days. Because validation and test do not agree on the best variant, I keep plain CSP rather than choose after seeing test results.

**Table 3.** The final network on four CSP variants. Test accuracies (%) were read after the final model was fixed; reservoir gain = ESN test − no-reservoir test.

| CSP variant | ESN val. acc. | ESN val. MSE | ESN test | No reservoir test | Reservoir gain |
|---|---|---|---|---|---|
| Plain CSP (final) | 82.0 | 0.548 | 78.3 | 77.4 | +0.8 |
| Fixed shrinkage 0.05 | 82.5 | 0.550 | 79.6 | 79.2 | +0.4 |
| Ledoit-Wolf [8] | 82.2 | 0.543 | 78.4 | 78.3 | +0.1 |
| Stationary CSP, weight 0.03 [9] | 83.2 | 0.548 | 76.7 | 75.6 | +1.1 |

### 4.5 Accuracy against parameters and the design history

Figure 7 places every model of the final family and both baselines by trained parameters. HWNet has about 408 times fewer trained parameters than MixNet for the same accuracy; the grey controls without a reservoir are smaller still (38 parameters) and only slightly less accurate. Table 4 lists the designs in the order I built them; the change that mattered most was replacing the gradient-trained readout by closed-form ridge regression with CMA-ES settings.

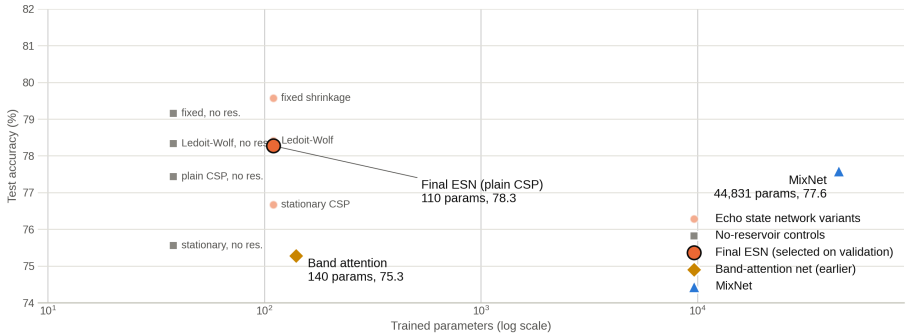



**Figure 7.** Session E accuracy against trained parameters (log scale). Orange: ESN with the ridge readout on each CSP variant; grey: the same without reservoir. Read after selection.

**Table 4.** Designs built during the project, in order (session E accuracy, read after each design was fixed).

| Design | Trained params | Test accuracy (%) |
|---|---|---|
| Band-attention network on windowed CSP features | 140 | 75.3 |
| ESN, gradient readout, MixNet's signals | 70 | 75.2 |
| ESN, gradient readout, shrinkage 0.05 + per-trial norm. | 70 | 76.8 |
| ESN, gradient readout, Ledoit-Wolf + per-trial norm. | 70 | 76.3 |
| ESN, ridge readout + CMA-ES, with direct inputs | 138 | 78.4 |
| **Final: ESN, ridge readout + CMA-ES, no direct inputs** | **110** | **78.3** |

### 4.6 Where the errors fall

Figure 8 pools the session E predictions of all subjects and folds (6,480 trials per model). The three models make errors of similar size in both classes: HWNet recognises 75.6 % of left-hand and 81.0 % of right-hand trials, MixNet 79.3 % and 75.9 %, FBCSP+SVM 76.9 % and 79.7 %. HWNet predicts "right hand" for 52.7 % of the trials, a slight lean that MixNet (48.3 %) does not share.

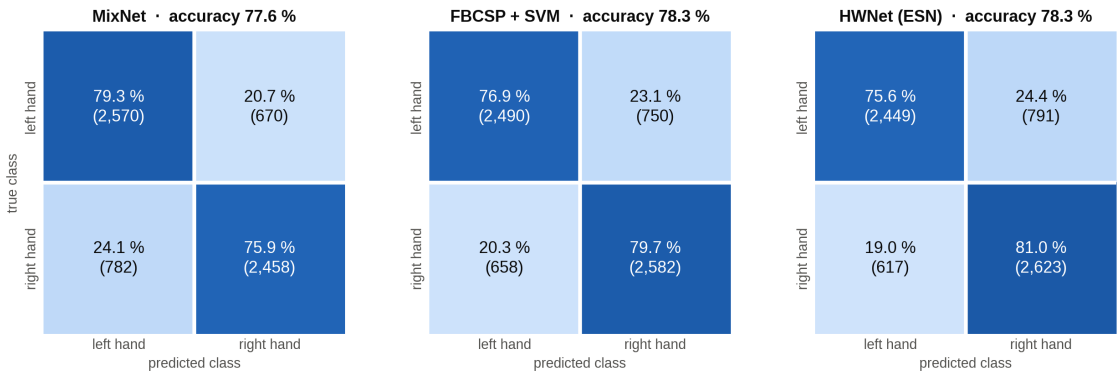



**Figure 8.** Confusion matrices on session E, pooled over 9 subjects and 5 folds. Rows sum to 100 %; counts in brackets.

Within single subjects the picture changes (Figure 9). The errors are often one-sided, with the direction depending on the subject: A04 recognises only 41 % of left-hand but 84 % of right-hand trials, A03 never misclassifies a right-hand trial but sends 18 % of left-hand trials to "right", and A05 leans the other way (82 % against 69 %). The mean gap between the two recalls of a subject is 15.6 points for HWNet, 13.6 for FBCSP+SVM and 19.8 for MixNet, so one-sidedness is common to all three models, not a property of the reservoir. Opposite leans in different subjects cancel in the pooled matrix. This is what a shift of each subject's decision threshold between the two days would produce; I did not test that explanation, and a label-free per-session correction is a natural next experiment.




**Figure 9.** HWNet confusion matrix of each subject on session E (5 folds). Rows sum to 100 %; counts in brackets; the title gives the subject's accuracy.

## 5 Discussion

**H1 is supported.** With 110 trained parameters, about 408 times fewer than MixNet trains, HWNet is level with MixNet and with FBCSP+SVM. The tests cannot show equivalence with nine subjects, but the means and the per-subject pattern are close.

**H2 is supported only weakly.** The reservoir adds 0.8 test points and 1.9 validation points to the same readout without it, and the gain has the same sign for all four CSP variants, but on test it is of the order of the spread between random reservoirs (0.4 points). The search chose a small input scale, so the reservoir works in the almost linear range of tanh and acts as a bank of random linear filters (Eq. 9); that would explain why a recurrent non-linear network gives so little over the filter bank and CSP that precede it. With the earlier gradient-trained readout and hand-picked settings the reservoir lost accuracy, so how the readout is trained matters as much as the reservoir.

**H3 is supported by two comparisons.** Per-trial normalisation helps by 2.2 test points and much less on validation, and HWNet loses 3.7 points from the training day to the test day against 6.7 for MixNet. Both observations rest on nine subjects. The confusion matrices add a caution: the day-to-day change appears to act on the decision threshold of each subject, which the per-trial normalisation does not address.

**Limitations.** Nine subjects give the paired tests little power. One readout per fold; CMA-ES optimised validation error averaged over three reservoirs, which still leaves room for fitting validation noise, and the input scale reached the edge of its search range. The no-reservoir controls reuse the ridge strength found for the reservoir model instead of their own. The module ablations remove one module at a time and do not test interactions. Several decisions (plain CSP, dropping the direct inputs) follow principle or validation where test results differ, and they are stated as such.

**Next steps.** A label-free per-session threshold or covariance correction for the one-sided errors; a readout over time windows, which would let the reservoir's memory contribute more than pooled power; a search that rewards small reservoirs and extends the input-scale range.

## 6 Conclusion

An echo state network with 54 random units and a 110-parameter ridge readout, on filter-bank CSP signals normalised within each trial, reaches 78.3 % on BCI Competition IV-2a, against 77.6 % for MixNet and 78.3 % for FBCSP+SVM, without a significant difference. Per-trial normalisation and standardisation contribute most among the new modules; the random reservoir adds a small gain that the data cannot separate clearly from the choice of reservoir. The errors of all models are one-sided within subjects, which points to the change of recording day as the remaining difficulty.

## Procedure, step by step

1. **Setup and baselines.** Cloned the course repository on branch `homework`, built a Python 3.8.10 virtualenv with TensorFlow 2.7.0, downloaded BCIC2a, reproduced MixNet (77.59 %) and ran the repository FBCSP+SVM (78.27 %).
2. **Earlier designs** (Table 4) are kept in `archive/`.
3. **Preprocessing.** Implemented filter-bank CSP, per-trial band normalisation and unit scaling in `HW_prep.py`, with optional shrinkage, Ledoit-Wolf and stationary CSP; wrote 5 leakage tests (all pass).
4. **Model.** Implemented the echo state network with a closed-form ridge readout in `HWNet.py`.
5. **Search.** Ran `cmaes_search.py` (CMA-ES, 300 settings, validation folds only) twice, with and without direct inputs, and fixed the final configuration by the rule stated in Section 2.5.
6. **Final run and benchmark.** Trained the final model into `logs/` and ran `benchmark.py --baseline MixNet --candidate HWNet`.
7. **Comparisons.** Trained the module ablations, two more random reservoirs and the CSP variants; read their test accuracies after step 6.
8. **Report.** Drew the figures with `make_figures.py`, `plot_module_outputs.py` and `plot_confusion.py`, and wrote this report from the saved result files.

## Reproducibility

Python 3.8.10, TensorFlow 2.7.0, cma 4.5.0, one RTX 3050 Laptop GPU; commands run from `experiments/` on branch `homework`. The final configuration is `rn_final` in `configs/HWNet.py`.

```
python download_datasets.py --dataset 'BCIC2a'
python prep_spectral_spatial_signals.py --dataset 'BCIC2a' --setting 'dependent'
python run_MixNet.py --model_name 'MixNet' --dataset 'BCIC2a' --train_type 'subject_dependent' \
    --data_type 'spectral_spatial_signals' --adaptive_gradient True \
    --policy 'HistoricalTangentSlope' --log_dir 'logs' --num_class 2 --GPU 0 \
    --margin 1.0 --n_component 2 --warmup 7
python prep_FBCSP.py --dataset BCIC2a --setting dependent
python run_FBCSP_SVM.py --dataset BCIC2a --data_type fbcsp --log_dir logs_compare
python prep_HW.py && python test_HW_leakage.py   # HW_PREP=<name> for other inputs
python cmaes_search.py                             # validation-only search
python run_HWNet.py && python benchmark.py --baseline MixNet --candidate HWNet
./run_variants.sh <variant> ...                  # ablations and CSP variants
python make_figures.py && python plot_module_outputs.py && python plot_confusion.py
```

## References


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[12] Course repository and assignment, https://github.com/xydxdy/MixNet, branch *homework*. The training loop, data loader, benchmark script and the FBCSP+SVM baseline are used unchanged.